

Biswajit Jena

Full Stack & Generative AI Engineer

"Applying First-Principles Thinking to solve real-world software challenges"

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TECHNICAL SKILLS

AI Engineering: LangGraph, LangChain, Vector DBs (Chroma)

Languages: Python, JavaScript (ES6+), HTML/CSS, Bash/Shell

Frameworks: Next.js, React, Node.js, Express.js,

Infrastructure: AWS (EC2), Linux Administration, Docker, Git, Nginx

PROJECTS

InstaGlowUp | *Next.js, MongoDB, AWS EC2* | [Source Code Link](#)

- Deployed a production-ready social platform on **AWS EC2** (Ubuntu), manually configuring Linux firewalls and environment secrets.
- Engineered a solution for server resource constraints by implementing **4GB Linux Swap Memory**, preventing build crashes on micro-instances.
- Integrated "Ghibli-style" generative art filters and automated content moderation pipelines using external AI APIs.

Autonomous Travel Agent | *LangGraph, Llama 3* | [Source Code Link](#)

- Developed an autonomous agent using a **cyclic graph architecture** in LangGraph for complex state management.
- Implemented the **ReAct pattern**, enabling the LLM to dynamically call external APIs (Tavily) to verify travel data.
- Reduced hallucination rates by enforcing a strict "Tool-First" logic flow, ensuring all itinerary output is backed by real-time search data.

Linux File MCP Server | *Python, MCP SDK* | [Source Code Link](#)

- Engineered a custom **Model Context Protocol (MCP)** server to facilitate secure communication between local LLMs and the Linux Kernel.
- Implemented a sandboxed execution environment allowing AI models to perform File I/O (Read/Write) without risking system integrity.
- Standardized the protocol for AI-System interaction, creating a reusable interface for local-LLM file management.

Marcus Hybrid RAG | *ChromaDB, Tavily* | [Source Code Link](#)

- Designed a **Hybrid RAG architecture** combining semantic vector retrieval (ChromaDB) with live web search.
- Synthesized structured historical datasets with unstructured live news, allowing the AI to apply philosophical principles to modern events.
- Optimized inference speed by utilizing **Groq's LPU** (Language Processing Unit) for near-instantaneous token generation.

CERTIFICATIONS

IBM Full Stack Software Developer Professional Certificate (Coursera)

Issued: Dec 2025 | [Verification Link](#)

EDUCATION

Berhampur University

Bachelor of Science in Physics (Honours)

Odisha, India

2022 – 2025